UTM

Universal Triage Model

universal approach for supporting industrial workflows

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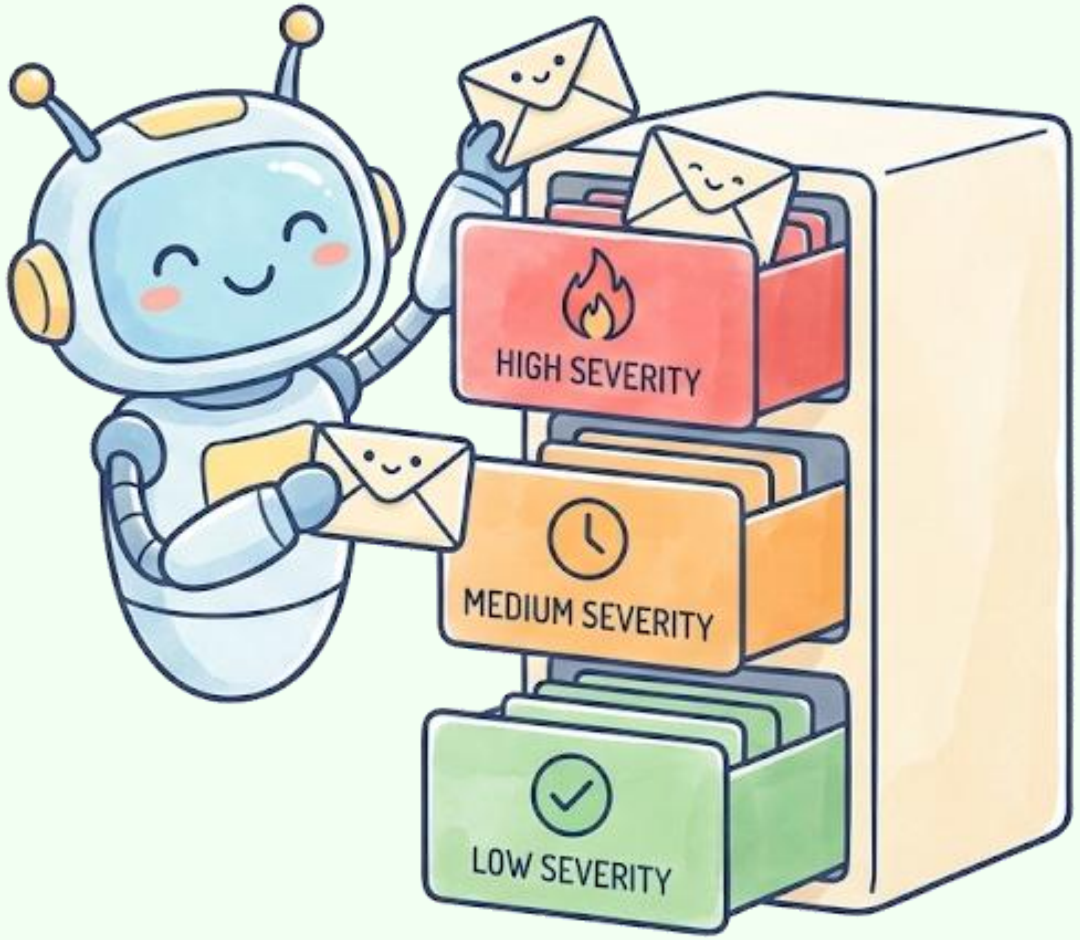


TECHNICAL APPROACH

Model Architecture

Model Pipeline

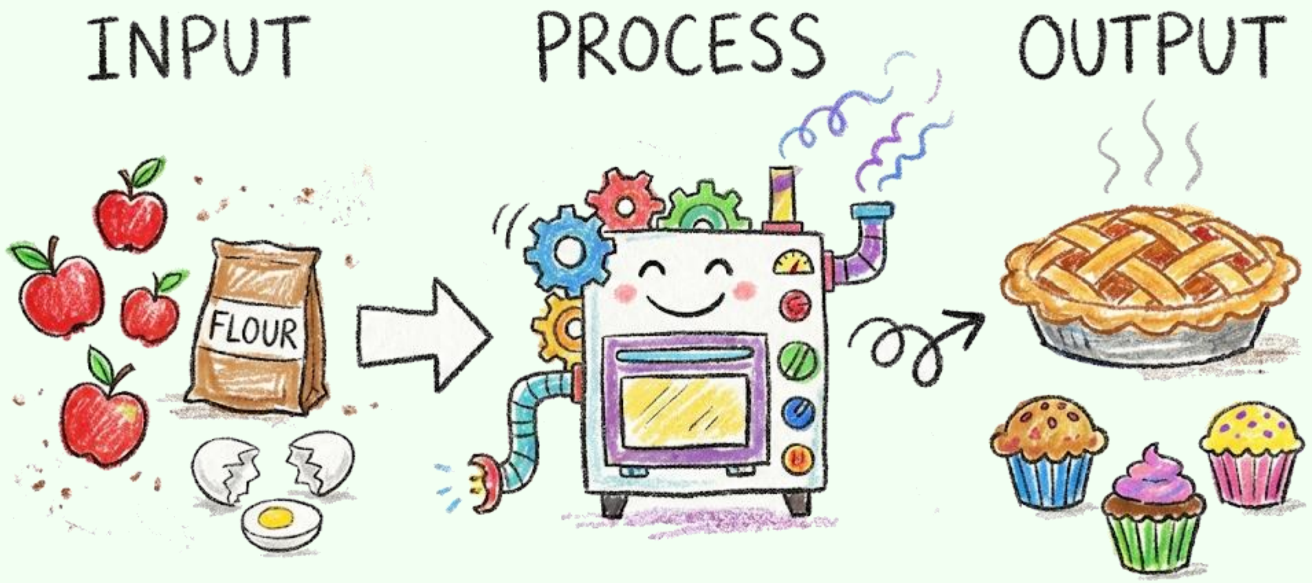
GPT-3.5 for data generation, Llama 3.2 baseline, RoBERTa fine-tuning

Input Processing

- Company background (industry, enterprise information system)
- Workflow description
- Components
- User report ticket

Output

- Affected component
- Severity level (Medium, High, Critical)

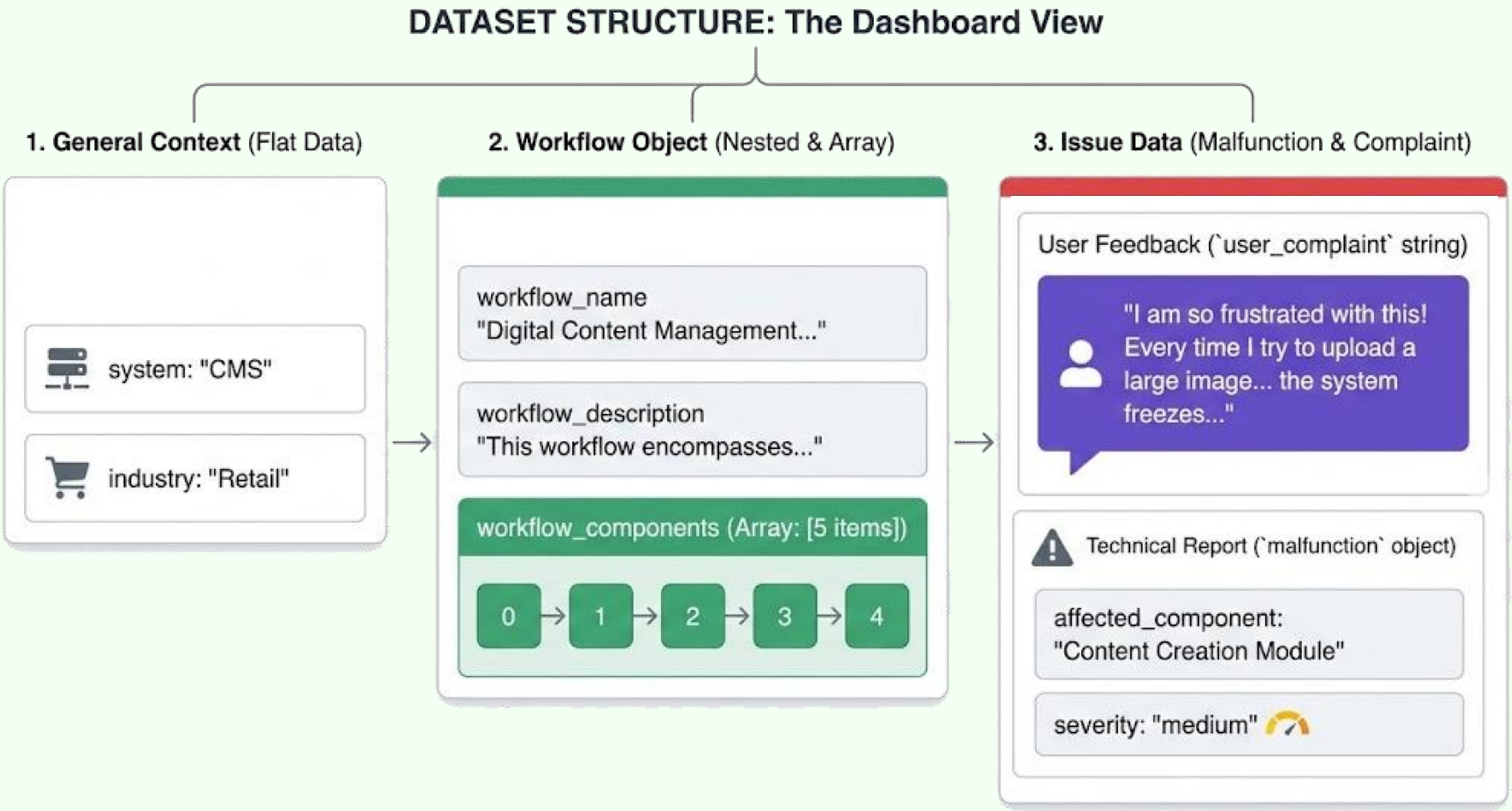


Component Metrics

Accuracy, Top-k ranking

Severity Metrics

F1 Score (macro), Accuracy, Weighted F1


DATASET STRUCTURE: The Dashboard View
1. General Context (Flat Data)
system: "CMS"
industry: "Retail"
2. Workflow Object (Nested & Array)
workflow_name
"Digital Content Management..."
workflow_description
"This workflow encompasses..."
workflow_components (Array: [5 items])
0
1
2
3
4
3. Issue Data (Malfunction & Complaint)
User Feedback (`user_complaint` string)
"I am so frustrated with this! Every time I try to upload a large image... the system freezes..."
Technical Report (`malfunction` object)
affected_component:
"Content Creation Module"
severity: "medium"

PROJECT OVERVIEW

Problem & Solution

Current Challenge

Manual ticket classification is costly and time-consuming

- High operational costs
- Slow response times

Our Innovation

Universal AI model adaptable across varied software systems

- Automated classification
- Multi-company deployment

TECHNICAL APPROACH

Model Development & Data Strategy

Data Generation

~2220 training examples, ~400 test examples

Baseline Models

Llama 3.2 for zero-shot and few-shot testing

Fine-Tuned Models

Component Classification

- Binary relevance classifier & ranking model
- Metrics: F1, Accuracy, Recall, Precision

Severity Classification

- RoBERTa-base for Critical/High/Medium
- Metrics: F1 (macro & weighted), Accuracy, Recall, Precision

Refinement Efforts

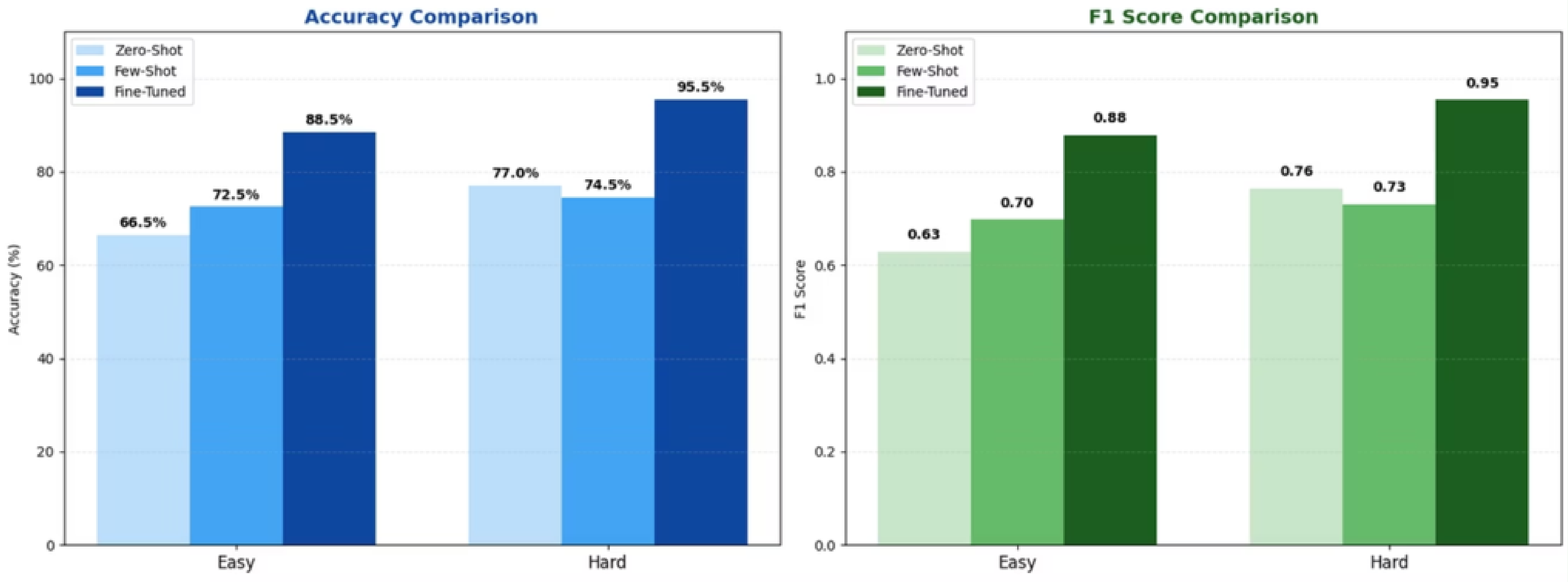
- Increase data diversity & quantity
- Adjust metrics (eval loss for best model)
- Component classification formulated as a ranking problem

Category	Component Model	Severity Model
Main Task	Binary Classification (Ranker Approach)	Multi-class Classification (3 severity levels)
Data Preparation	**Explosion:** Splitting each issue into separate rows for every component	**Contextual Prompt:** Including Industry, System, and User Experience
Number of Epochs	3	10 (with Early Stopping)
Learning Rate (LR)	3e-5	1e-5
Layer Freezing	None (Full training)	Freezing the first 4 layers
Imbalance Management	WeightedTrainer (weights for Label 1)	WeightedTrainer (weights for Critical and High)
Primary Metrics	**Top-2 Accuracy** and F1	**Macro-F1** and Accuracy
Regularization	Early Stopping, Weight Decay	Early Stopping, Weight Decay, Layer Freezing, Warmup

Results: Seen vs. Unseen Context

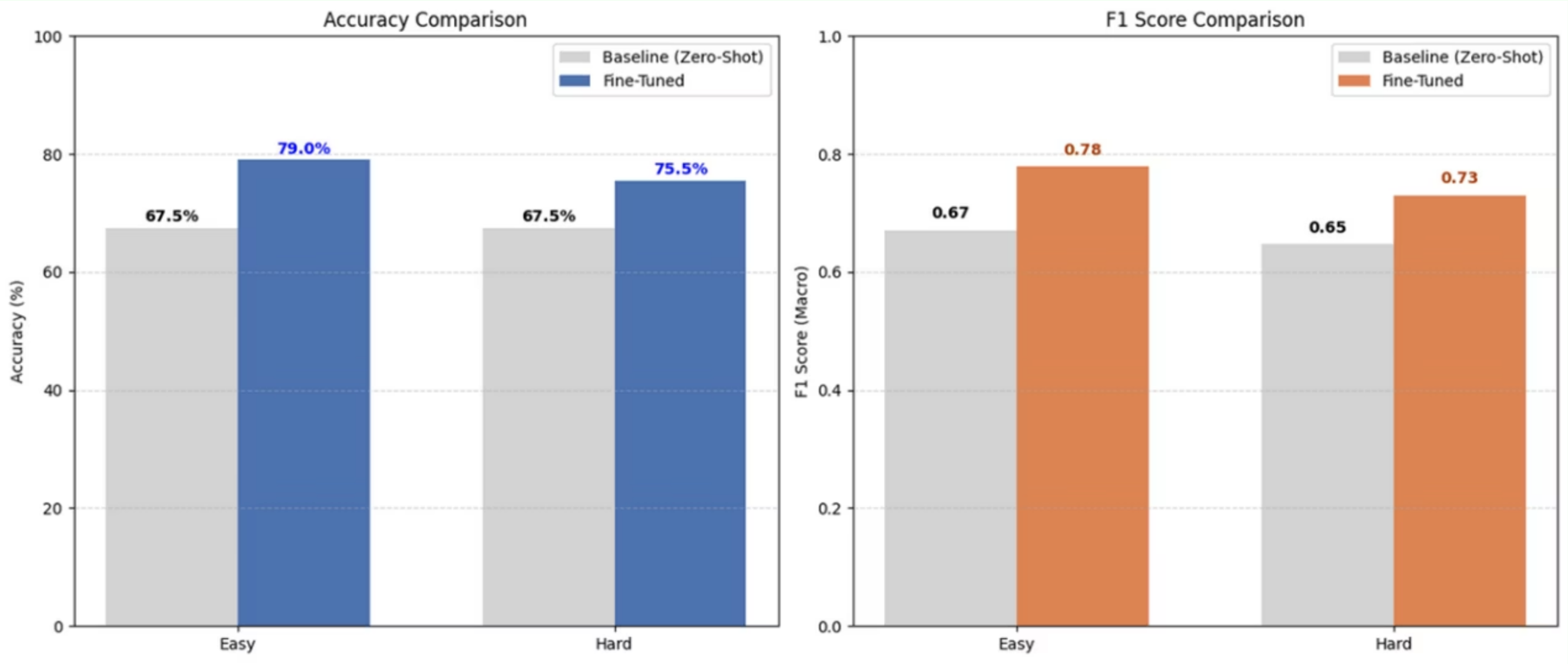
Component Classification Accuracy

Easy Set (Seen Context)	Hard Set (Unseen Context)
Tests the model on new tickets from workflows already encountered during training	Tests the model on entirely new companies and workflows that were not part of the training data.



Dataset	Zero-Shot	Few-Shot	Fine-Tuned	Gain
Easy	66.5%	72.5%	88.5%	+16.0%
Hard	76%	73%	95.5%	+19.5%

Severity Classification Performance



Dataset	Baseline Acc	Fine-Tuned Acc	Baseline F1	Fine-Tuned F1
Easy	67.5%	79%	0.6708	0.7783
Hard	67.5%	75.5%	0.6475	0.7303

INSIGHTS

Key Learnings & Future Directions

Achievement

Successfully trained universal model that predicts components and severity on unseen workflows

Critical Lessons

- Clear label definitions essential
- Testing on unseen data is crucial

Future Exploration

Real-World Data

Deploy with multiple companies, collect real tickets for enhanced training

Active Learning

User corrections as supervision signals for continuous model improvement